

Inferential statistics

Dr Sandheep Sugathan

MBBS, DPH, MD, FAIMER Fellow

Senior Lecturer in Public Health

University Kuala Lumpur Royal College of Medicine Perak

INFERENCE STATISTICS

- Inferential statistics or statistical inference is the process of applying statistical methods in order to draw conclusions from sets of data that arise from systems which are affected by random variation.
- Otherwise known as Analytical statistics.

Inferential statistics

- Inferential statistics makes inferences about populations using data drawn from the population.
- Inference or Prediction of association between a dependent and an independent variable is done
- Prediction of difference of a dependent variable between 2 groups is done
- Prediction of association between 2 or more variables in different groups is done

Variable

A variable is an element, feature, or factor that is liable to vary or change.

Variables in a study can be dependent variables or independent variables.

Dependent variable

Dependent variable is a factor or phenomenon that is changed by the effect of an associated factor or phenomenon called the independent variable.

It is a variable whose value depends on that of another variable.

Independent variable

The independent variable is typically the variable which is manipulated or changed and the dependent variable is the variable which is affected by the changes in the independent variable.

For example concerning nutrition, the independent variable of daily vitamin C intake (how much vitamin C one consumes) can influence the dependent variable of immunity status of the patient.

Regular physical activity can affect the variable **obesity**. So in this scenario regular physical activity is the independent variable and obesity is the dependent variable.

Obesity can lead to the development of **Type 2 Diabetes mellitus**. So in this scenario, obesity is the independent variable and development of Type 2 Diabetes mellitus is the dependent variable.

Difference of occurrence of disease between two categories can be **real difference** or **due to chance**.

Tests of significance are applied to know if the difference between two categories is statistically significant or not.

Inferential statistics – uses application of various statistical tests of significance

Inferential statistics checks if there is

- A **real association** between a dependent and an independent variable that means there is a **statistically significant association** between dependent and independent variables.
- A **real difference** in dependent variable in 2 or more groups that means there is a **statistically significant difference** in dependent variable in 2 or more groups.

Tests of statistical significance are applied to know if the difference between mean values in 2 or more groups, the difference between 2 proportions in 2 or more groups or the difference in occurrence of disease or a problem between 2 categories is statistically significant.

Inferential statistics is done by using either

- Parametric or Non parametric tests of statistical significance
- Correlation analysis
- Linear regression analysis
- Calculation of 95% confidence intervals of difference between means in different groups
- Calculation of 95% confidence intervals of difference between proportions in different groups
- Logistic regression analysis

If there is a **real association or difference**, the statistical test for significance shows significance.

If there is no real association or difference, the statistical test for significance shows no significance.

P value is very important in Inferential statistics

P-value or Significance value is the **probability to have an association between the dependent variable and the independent variable due to chance**. It is the probability to have the difference between dependent variables in two different groups due to chance alone.

Errors in a study

	H_0 is true	H_0 is false
accept H_0	correct	type II error = β
reject H_0	type I error = α	correct power = $1 - \beta$

Type 1 error or α error (Null hypothesis is true but it is rejected) is fixed at 5% or less (p value is dependent on α error.) Type 1 error is more dangerous.

Type 2 error or β error (Null hypothesis is false but it is accepted) is fixed at 20% or less

$(1 - \beta)$ is known as power of the study and it should be 80% or more

Before applying the appropriate statistical test, the researcher need to assume that there is no association between the dependent and independent variables or there is no difference between different groups.

Null Hypothesis

In inferential statistics, the term "null hypothesis" is a **general statement or assumption that there is no relationship or association between dependent and independent variables or no difference between dependent variable in 2 different groups. If there is any difference or association seen, it is due to chance.**

It also states that if any association or difference is observed between the variables or groups, it is due to chance alone or due to sampling error. It is denoted by **H_0** ("H-null" or "H-zero").

Statistical test of significance is applied to prove or disprove the null hypothesis.

Alternative Hypothesis

Alternative hypothesis or the research hypothesis is a general statement or assumption that there is a relationship or association between dependent and independent variables or difference between dependent variable in 2 different groups.

Then appropriate **Parametric or Non parametric test of statistical significance is applied**

Parametric tests of Statistical significance

A parametric statistical test is one that makes assumptions about the parameters (defining properties) of the population distribution(s) from which one's data are drawn.

Typical **assumptions** which need to be true **for applying parametric tests** are:

- a. Normality of data:** Data should show have a normal distribution
- b. Homogeneity of variances of data:** Data from multiple groups should have the similar variance
- c. Linearity of data:** Data have a linear relationship between each other. That means data should be collected from a homogeneous population
- d. Independence of data:** Data are independent and not biased using a probability sampling method

First test is for assumptions for selection of appropriate statistical test

Normality test – Normality of data is tested using

- **Visual methods** – Histogram / Normal distribution curve / Box and Whisker plot / QQ Plot

OR

- **Shapiro Wilk Test** – if sample size is less than 40
- **Kolmogorov Smirnov Test**

If data is normally distributed – Select **parametric tests of statistical significance**

Shapiro Wilk Test is the test of normality

If Significance value is >0.05 , data is normally distributed

If Significance value is <0.05 , data is not normally distributed

If data is normally distributed – select Parametric tests

Commonly used **Parametric tests** of Statistical significance

- Independent Samples T test
- Paired T test
- One sample T test
- ANOVA test (Analysis of variance)
- Pearson Correlation test
- Linear regression test

If data is **not normally distributed** – Select **non parametric tests** of statistical significance

Commonly used Non Parametric tests of Statistical significance

- Chi square test
- Chi square test with Fisher exact test
- Wilcoxon Signed Rank test
- Mann Whitney U test
- Kruskal Wallis test
- Spearman Correlation test

If P-value or Significance value in the parametric or non-parametric statistical test is less than 5% or less than 0.05,

the probability to have an association between dependent variable and independent variable due to chance alone is less than 5% or less than 0.05.

If P-value is less than 5% or less than 0.05, the association is real and not due to chance.

That means that the association or difference is **statistically significant**

If P-value or Significance value in the parametric or non-parametric statistical test is more than 5% or more than 0.05,

the probability to have an association between dependent variable and independent variable due to chance alone is more than 5% or more than 0.05.

If P-value is more than 5% or more than 0.05, the association is considered not real and is due to chance alone.

That means the association is **not statistically significant**

PARAMETRIC TESTS OF STATISTICAL SIGNIFICANCE

Scenario	A research was done to know if the mean haemoglobin value among the patients was significantly different from the standard haemoglobin value which was obtained from the guidelines. The test of normality was done and it showed that data was normally distributed in the sample population.
Dependent variable	Mean haemoglobin value in the sample (continuous variable – interval scale)
Independent variable	standard haemoglobin value
Test of statistical significance	One sample T test
Justification	- T test because data is normally distributed (T test is a parametric test. T test is applied for comparison of 2 means.) - One samples T test – because mean haemoglobin in the sample population is compared with the standard haemoglobin value which is obtained from the guidelines.

Scenario	A research was done to know if the mean systolic blood pressure value among the patients was significantly different from the standard systolic blood pressure value which was obtained from the clinical practice guidelines. The test of normality was done and it showed that data was normally distributed in the sample population.
Dependent variable	Mean systolic blood pressure value in the sample (continuous variable – interval scale)
Independent variable	standard systolic blood pressure value which was obtained from the clinical practice guidelines
Test of statistical significance	One sample T test
Justification	- T test because data is normally distributed (T test is a parametric test. T test is applied for comparison of 2 means.) - One samples T test – because mean systolic blood pressure in the sample population is compared with the standard systolic blood pressure value which is obtained from the guidelines.

Scenario	A research was done to know if the mean haemoglobin value among males was significantly different from mean haemoglobin value among females. The test of normality was done and it showed that data is normally distributed in the sample population among both genders.
Dependent variable	Haemoglobin values (continuous variable – interval scale)
Independent variable	Gender – categorical variable
Test of statistical significance	Independent samples T test
Justification	- T test because data is normally distributed (T test is a parametric test. T test is applied for comparison of means in 2 groups) - Independent samples T test – because means in 2 independent groups or unrelated groups are compared. Here males and females are 2 independent or unrelated groups

Scenario	A research was done to know if the mean knowledge scores among employed females was significantly different from mean knowledge scores among unemployed females. The test of normality was done and it showed that data was normally distributed in the sample population among both the groups under study.
Dependent variable	Knowledge scores (continuous variable)
Independent variable	Occupation level (categorical) – Employed or unemployed
Test of statistical significance	Independent samples T test
Justification	- T test because data is normally distributed (T test is a parametric test. T test is applied for comparison of means in 2 groups) - Independent samples T test – because means in 2 independent groups or unrelated groups are compared. Here employed females and unemployed females are 2 independent or unrelated groups

Scenario	A research was done to know if the mean haemoglobin value among 100 pregnant women before starting treatment with iron tablets for 3 months period was significantly different from mean haemoglobin value after treatment. The test of normality (Shapiro Wilk test) was done and it showed that data was normally distributed in the sample.
Dependent variable	Haemoglobin value in sample population (continuous variable)
Independent variable	treatment with iron tablets for 3 months period
Test of statistical significance	Paired sample T test
Justification	- T test because data is normally distributed (T test is a parametric test. T test is applied for comparison of means in 2 groups) - Paired sample T test – because mean haemoglobin value in the sample before treatment with iron tablets for 3 months period is compared with the mean haemoglobin value in the sample after treatment. Both the values (are from the same set of patients). So it is a paired sample

Scenario	A research was done to know if the mean knowledge scores among 100 diabetic patients before a health education intervention was significantly different from mean knowledge scores among 100 diabetic patients after the health education intervention. The test of normality (Shapiro Wilk test) was done and it showed that data (knowledge scores before and after the health education intervention) was normally distributed in the sample.
Dependent variable	knowledge scores among patients (continuous variable)
Independent variable	Attending a health education intervention
Test of statistical significance	Paired sample T test
Justification	- T test because data is normally distributed (T test is a parametric test. T test is applied for comparison of means in 2 groups) - Paired sample T test – because mean knowledge scores in the sample before health education intervention is compared with mean knowledge scores in the sample after health education intervention. Both the values (are from the same set of patients). So it is a paired

Scenario	A research was carried out to know if the mean fasting blood sugar values among individuals from different ethnic groups (Malay, Chinese, Indians and others) were significantly different among each other. Shapiro Wilk test showed that the data was normally distributed in the sample.
Dependent variable	fasting blood sugar values among individuals (continuous variable)
Independent variable	Ethnicity (categorical variable with more than 2 groups)
Test of statistical significance	ANOVA (Analysis of Variance) Test
Justification	- ANOVA test because data is normally distributed (ANOVA is a parametric test and is applied for comparison of means in more than 2 independent groups) - Here mean FBS of individuals from different ethnic groups are compared (There are more than 2 groups).

Scenario	A research was carried out to know if the mean haemoglobin values among individuals from different socio economic class (low, middle and high) were significantly different among each other. Shapiro Wilk test showed that the data was normally distributed in the sample.
Dependent variable	Haemoglobin values among individuals (continuous variable)
Independent variable	Socio economic class (categorical variable with more than 2 groups)
Test of statistical significance	ANOVA (Analysis of Variance) Test
Justification	- ANOVA test because data is normally distributed (ANOVA is a parametric test and is applied for comparison of means in more than 2 independent groups) - Here mean haemoglobin of individuals from different socio economic class are compared (There are more than 2 groups).

NON PARAMETRIC TESTS OF STATISTICAL SIGNIFICANCE

Chi-square test

It is the Non parametric test of significance which is applied to know the association between two categorical variables.

It can be applied to know if the difference in the prevalence or proportion of an outcome in different groups is statistically significant.

Scenario	A research was conducted to know if the prevalence or proportion of hypertension among males and females in a sample population was significantly different.
Dependent variable	Hypertension – prevalence or proportion of hypertension – Categorical variable
Independent variable	Gender – categorical variable
Test of statistical significance	Chi-square test
Justification	Chi-square test is the appropriate statistical test - To study the association between two categorical variables (Here Hypertension and gender are categorical variables) - To study the difference between proportions in 2 or more different groups (Here to study the proportion / prevalence of hypertension in two different groups (two genders))

Scenario	A study was carried out to know if the prevalence of diabetes mellitus among obese and non-obese individuals were significantly different. Which test of significance is appropriate in the above situation?
Dependent variable	Prevalence or proportion or occurrence of diabetes mellitus – Categorical variable
Independent variable	Obesity – categorical variable – obese or non-obese
Test of statistical significance	Chi-square test
Justification	Chi-square test is the appropriate statistical test - To study the association between two categorical variables (Here prevalence of diabetes mellitus and obesity are categorical variables) - To study the difference between proportions in 2 or more different groups (Here to study the proportion / prevalence of diabetes mellitus in two different groups (obese and non-obese))

Scenario	A research was done to know if the mean fasting blood sugar value among 100 diabetic patients before starting treatment with anti diabetic medication for 6 months period was significantly different from mean fasting blood sugar value after treatment. The test of normality (Shapiro Wilk test) was done and it showed that data was not normally distributed in the sample.
Dependent variable	Fasting blood sugar value in sample population (continuous variable)
Independent variable	treatment with anti diabetic medication for 6 months period
Test of statistical significance	Non parametric test (Wilcoxon Signed Rank test / Mann Whitney U test)
Justification	- Non parametric test because data is not normally distributed - Comparison of means of related samples – because mean fasting blood sugar value in the sample before treatment with anti diabetic medication for 6 months period is compared with the mean fasting blood sugar value in the sample after treatment. Both the values are from the same set of patients. So it is a related sample

Scenario	A research was done to know if the mean knowledge scores among educated individuals was significantly different from mean knowledge scores among uneducated individuals. The test of normality was done and it showed that data was not normally distributed in the sample population among both the groups under study.
Dependent variable	Knowledge scores (continuous variable)
Independent variable	Education level (categorical) – Educated or uneducated
Test of statistical significance	Non parametric test for comparing two independent groups (Wilcoxon Signed Rank test / Mann Whitney U test)
Justification	- Non parametric test because data is not normally distributed - Comparison of unrelated samples was done – because 2 independent groups or unrelated groups are compared. Here educated and uneducated individuals are 2 independent or unrelated groups

Scenario	A research was carried out to know if the mean systolic blood pressure values among different occupation groups (Professionals, Clerical staff, House keeping staff) were significantly different among each other. Shapiro Wilk test showed that data was not normally distributed.
Dependent variable	systolic blood pressure values (continuous variable)
Independent variable	Occupation categories (categorical variable with more than 2 groups)
Test of statistical significance	Kruskal Wallis Test
Justification	- Kruskal Wallis test because data is not normally distributed (Kruskal Wallis is a non parametric test and is applied for comparison of means in more than 2 independent groups when data is not normally distributed) - Here mean systolic blood pressure values of individuals from different occupation groups are compared (There are more than 2 groups)

**Inferential statistics
using 95% Confidence intervals**

Sample statistic – is the statistical value that we get in a study sample.

It can be Sample mean or Sample proportion.

If mean or proportions that we obtain from all the possible samples from a population are plotted as a graph, it shows a normal distribution.

All the sample means / sample proportions are normally distributed around the **true population mean / true population proportion**

STANDARD ERROR OF MEAN or SAMPLING ERROR

shows how the sample mean deviates from the true population mean. It can be calculated by

$$= \frac{SD}{\sqrt{n}}$$

SD is the standard deviation and n is the sample size.

In 95% times, Population mean (true value in the population) will be within the range of sample mean ± 1.96 times of Standard error of mean.

The range between (mean – 1.96 times standard error of mean) and (mean + 1.96 times standard error of mean) is known as **95% confidence limits or 95% confidence intervals of mean.**

95% of times, true population mean will be within the range of 95% confidence limits.

If mean hemoglobin of 100 anemic patients is 11.8 g/dl and standard deviation is 4

$$\begin{aligned}\text{Standard error of mean} &= \text{SD} / \sqrt{n} \\ &= 4 / \sqrt{100} \\ &= 4 / 10 \\ &= 0.4\end{aligned}$$

95% Confidence Intervals of mean

$$\begin{aligned}&= \text{mean} \pm 1.96 \times 0.4 \\ &= 11.8 \pm 0.784 \\ &= 11.016 \text{ to } 12.584\end{aligned}$$

If you repeat the research 100 times, 95% of times, true population mean will be within the range of 95% confidence limits.

That is between 11.016 and 12.584

In another terms we can be 95% confident that the true population mean will be between 11.016 and 12.584

STANDARD ERROR OF PROPORTION shows how the sample proportion deviates from the population proportion.

$$= \sqrt{\{ [P \times (100 - P)] / n \}}$$

P = sample proportion, n = sample size

In 95% times, Population proportion will be within the range of sample proportion ± 1.96 times of Standard error of proportion.

The range between (proportion – 1.96 times standard error of proportion) and (proportion + 1.96 times standard error of proportion) is known as **95% confidence limits or 95% confidence intervals of proportion.**

95% of times, population proportion will be within 95% confidence limits of proportion.

If proportion of GDM among 100 pregnant women is 20 %

$$\begin{aligned}\text{Standard error of proportion} &= \sqrt{\{ [P \times (100 - P)] / n \}} \\ &= \sqrt{\{ [20 \times (100 - 20)] / 100 \}} \\ &= \sqrt{\{ [20 \times 80] / 100 \}} \\ &= \sqrt{\{ 1600 / 100 \}} = \sqrt{16} = 4\end{aligned}$$

95% Confidence Intervals of proportion

= sample proportion $\pm$ 1.96 X standard error of proportion

= 20 $\pm$ 1.96 X 4 = 20 $\pm$ 7.84

= 12.16 to 27.84

95% of times, true population proportion of GDM will be within 12.16 to 27.84 %

STANDARD ERROR OF DIFFERENCE BETWEEN MEANS in 2 different samples is

$$\sqrt{\frac{SD_1^2}{n_1} + \frac{SD_2^2}{n_2}}$$

SD_1 = Standard deviation of variable in the first study sample

n_1 = Sample size in the first study sample

SD_2 = Standard deviation of variable in the second study sample

n_2 = Sample size in the second study sample

The range between (differences between means – 1.96 times standard error of difference between means) and (difference between means + 1.96 times standard error of difference between means) is known as 95% confidence limits or 95% confidence intervals of difference between means.

If difference between means is given and the 95% confidence intervals of difference between means is also given

95% of times, difference between means will be within 95% confidence limits of difference between means.

If 95% confidence intervals of difference between means includes zero, the difference between means is not significant.

Mean Fasting Blood Sugar values of 2 groups of patients are given

Group 1 mean FBS = 5.8 mmol / l (n = 100)

Group 2 mean FBS = 6.2 mmol / l (n = 120)

Difference between mean = 0.4

95% confidence intervals of difference between means is given as (-0.1 to 0.9)

Difference between means is not statistically significant (because 95% CI includes zero)

STANDARD ERROR OF DIFFERENCE BETWEEN PROPORTIONS

$$\sqrt{\frac{P1 (100 - P1)}{n1} + \frac{P2 (100 - P2)}{n2}}$$

P1 = proportion of outcome in the first study sample, (100 – P1) is the proportion of no outcome in the first sample and n_1 = sample size in the first study sample

P2 = proportion in the second study sample, (100 – P2) is the proportion of no outcome in the second sample and n_2 = sample size in the first study sample

The range between

(*difference between proportions – 1.96 times standard error of difference between proportions*) and (*difference between proportions + 1.96 times standard error of difference between proportions*) is known as **95% confidence limits** or 95% confidence intervals of difference between proportions.

If difference between proportions is given and the 95% confidence intervals of difference is given

95% of times, difference between proportions will be within 95% confidence limits of difference between proportions.

If 95% confidence intervals of difference between proportions includes zero or touches zero, the difference between proportions is not significant.

Prevalence of diabetes among 2 categories of individuals are given

Group 1 (obese) 30% are diabetic (n = 100)

Group 2 (non obese) 15% are diabetic (n = 100)

Difference between 2 proportions = 15%

95% confidence intervals of difference between proportions is given as (3.632% to 26.368%)

Difference between proportions is statistically significant (95% CI of difference between proportions does not include zero)

If 95% confidence interval is wide

- Precision is less

If 95% confidence interval is narrow

- Precision is more

- So 95% CI is a measure of precision of the study